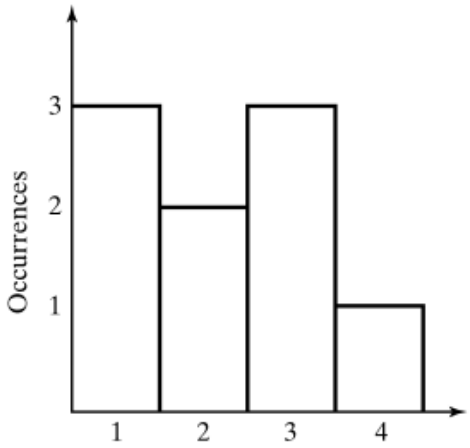
Useful Statistical Models

Queuing systems

- In queueing systems, such as those found in manufacturing or service industries, the timing of arrivals and the duration of services can vary.
- Sometimes these timings are predictable, like a conveyor belt moving at a constant speed in a factory, or a robotic arm performing spot welding consistently on a production line. However, in many cases, these timings are probabilistic, meaning they follow a probability distribution.
- Example: Mechanics arrive at a tool crib where attendants provide them with the necessary tools.
- The times between each mechanic's arrival are recorded. For instance, the first mechanic may arrive at time 0, the second at 7 minutes and 13 seconds, and so on.
- Alternatively, we can record the number of arrivals within fixed time intervals. For example, in the first 10-minute period, one mechanic arrives at 10:05:03, and in the second period, two mechanics arrive, and so forth.
- This data can be summarised in a histogram, showing the distribution of arrivals over time intervals.



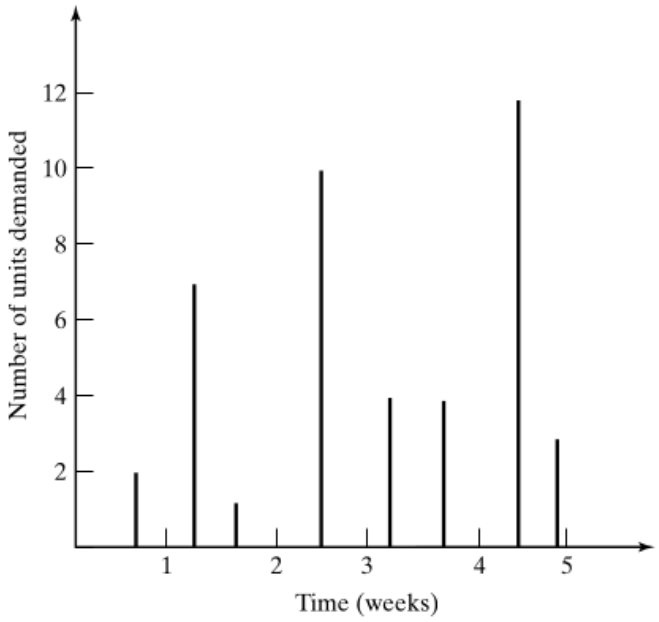
- We need an understanding of the distribution of arrival times and the number of arrivals per time for simulating waiting-line systems, which are prevalent in various settings like manufacturing plants, job shops, or call centres.
- Service times, on the other hand, can also vary. They may be constant, but random factors can introduce variability. In such cases, the *normal distribution* might describe the service time well.
- Sometimes, we encounter situations where a random variable follows a normal distribution but is constrained to be greater than or less than a certain value. In these cases, we use a *truncated normal distribution*.
- Additionally, we have other distributions like *gamma* and *Weibull*, which are commonly used to model both interarrival and service times.
- These distributions offer different shapes and tail behaviours compared to the *exponential* distribution, providing more flexibility in modelling real-world scenarios.
- For instance, the Weibull distribution can better capture scenarios with longer or shorter tails than the exponential distribution, making it a valuable tool in simulation modelling.

Inventory systems

- In inventory systems we typically deal with three key random variables:
    a. Number of Units Demanded: This refers to how many units are requested in each order or within a specific time frame.
    b. Time Between Demands: It represents the duration between consecutive demands.
    c. Lead Time: This is the time it takes from placing an order to receiving it.

- While simple inventory models might assume constant demand and zero or constant lead time, real-world scenarios are more complex.

- In practice, demands occur randomly over time, and the quantity demanded can vary each time.



- When it comes to modelling these uncertainties, we often rely on probability distributions. While theoretical models might use distributions for mathematical convenience, real-world scenarios might call for different assumptions. Simulation models offer flexibility in accommodating realistic assumptions.
- For example, the *gamma distribution* is commonly used to model lead times in inventory systems. It's found to be a good fit in many cases.
- Other distributions like geometric, Poisson, and negative binomial cater to various demand patterns:
    1. *Geometric Distribution*: This is a special case of the negative binomial and is useful when demand patterns have a mode at unity (meaning at least one demand occurs).
    2. *Negative Binomial Distribution*: It's suitable for scenarios with long-tail demand data.

3. *Poisson Distribution*: Often used due to its simplicity, extensive tabulation, and familiarity. It's good for modelling demand, especially when the tail of the distribution is shorter compared to the negative binomial.

Reliability Problems

- In reliability and maintainability analysis, the time to failure of components or systems is a critical factor. Various probability distributions are used to model this time:
    a. Exponential Distribution: Often used when only random failures occur. It assumes a constant failure rate over time.
    b. Gamma Distribution: Arises from modelling standby redundancy, where components have exponential time to failure.
    c. Weibull Distribution: Widely used for time to failure modelling due to its flexibility in approximating different phenomena. It's particularly effective when failures result from the most serious defects among many possibilities.
    d. Normal Distribution: Suitable for situations where failures are primarily due to wear and tear.
    e. Lognormal Distribution: Applicable in describing time to failure for certain types of components.

Limited Data

1. When data collection is incomplete or ongoing, simulations may need to start before all data is available. In such cases, three distributions are commonly used:
    a. Uniform Distribution: Used when the randomness of an event is known, but details about its distribution are lacking. Some critics dub it the "distribution of maximum ignorance" because it only specifies the continuous interval of possible values.

b. Triangular Distribution: Assumes minimum, maximum, and modal values of the random variable, providing a simple way to represent uncertainty.
c. Beta Distribution: Offers various forms on the unit interval and can be shifted to fit desired intervals. It encompasses the uniform distribution as a special case.